UNITED STATES FOREST SERVICE (USFS)
INCIDENT PROCUREMENT OPERATIONS (IPO)
REQUEST FOR INFORMATION (RFI)
LIDAR SYSTEM PAYLOAD

SUBMITTAL OF INFORMATION:

This RFI is for market research only and does not constitute a solicitation for proposals. The information provided may be utilized by the USFS in developing its acquisition strategy and requirements documents. It does not commit the Government to pay any cost incurred in the submission of responses or in making necessary studies or designs, nor to contract for supplies or services. No funds have been authorized, appropriated, or received for this contemplated effort. Any cost incurred shall be at the Offeror's own risk. Live demonstrations that are currently integrated with approved USFS unmanned aircraft may be requested at a future date. Responses shall include potential Offeror's Unique Entity Identifier (UEI), be in PDF format, and emailed to the Contracting Officer by 1200 hours MT, March 31, 2026, at steven.b.peterson@usda.gov.

DESCRIPTION:

The purpose of this notice is to obtain information regarding potential Offerors capable of providing Light Detection and Ranging (LiDAR) sensor systems for use with approved USFS agency fleet for the purpose of mapping and surveying our National Forests and Grasslands. The Government is seeking information regarding LiDAR payloads that are integrated with any of the following Unmanned Aircraft (UA):

- Freefly AltaX
- Freefly Astro
- Vision Aerial Switchblade

BACKGROUND:

The Government is interested in the use of LiDAR system payloads to map and survey targeted areas of interest with the use of Unmanned Aircraft System (UAS). This may include resource management activities (vegetation mapping and characterization, flood plain mapping, erosion monitoring, hazard/disaster assessment, terrain analysis, heritage site mapping, etc.) as well as engineering and survey activities (infrastructure inspections, topographic surveys for infrastructure development, etc.). Provide written documentation of the capability to perform UAS LiDAR data collection. Include UAS and LiDAR system payload specifications. If equipment does not appear to meet minimum agency mission needs or integrations to approved USFS agency fleet, you will not be considered to perform the demonstration, should they occur. If you cannot perform the tasks, provide a description and rationale for what it may take to integrate the payload to agency UA to accomplish the task (funding, additional time, additional resources such as multiple payloads, UAS, or pilots, etc.). The government is interested in UAS and payload configurations that could meet or exceed the requirements of this RFI. Any demonstration requests will require current FAA-qualified Remote Pilot(s) controlling the aircraft. Proof of qualification will be required before approval of any demonstration.

TECHNICAL REQUIREMENT:

LiDAR payloads will be used in conjunction with USFS approved UAS to map and survey the National Forests and Grasslands in a predictable and controlled manner in real-time settings. The UAS mounted device should be able to collect data for areas of interest whether preplanned or under full control by the operator to manipulate the results.

- LiDAR scanner with integrated Inertial Measurement Unit (IMU), GPS, and RGB Camera for direct georeferencing and colorization of LiDAR point cloud data
- All submissions will require "No Technical Objection" issued by the aircraft manufacturer for the payload/aircraft configuration
- Country of origin for sensor payload
- Integrated system with simple, limited steps for operation
- Maximum physical dimensions of 35 cm (length) x 20 cm (width) x 20 cm (height)
- Maximum weight of sensor must be in accordance with each individual platform's payload capacity and meet Part 107 requirements
- Operating temperature range -20 to +40 degrees C (-4 to +104 degrees F) or better
- Wavelength: 905 nm Class 1 eye safe laser
- Scanner precision 2 cm or less
- Scanner accuracy 3 cm or less
- IMU geospatial accuracy after post processing of 0.02 m horizontal and 0.05m vertical
- Horizontal Field of View (HFOV) + 50 to - 50 degrees (100 degrees total) or larger
- Vertical Field of View (VFOV) +15 to -15 degrees (30 degrees total) or larger
- Absolute geopositional accuracy up to 2-5 cm or better at 80 meters AGL
- Scan rate of 200 lines per second or more
- Range of 20% reflective surfaces at least 100 m
- Pulse repetition rate (PRR) up to 240,000 measurements per second or larger
- Up to 3 returns or echoes per pulse measurement or more
- 8-bit Intensity values provided for the signal strength of the returns / echoes
- User selectable pulse repetition rates (PRR) in units of kHz
- Laser beam divergence equal to or less than 0.28 degrees horizontal and 0.03 degree vertical
- Integrated RGB cameras for contemporaneous collection of imagery with LiDAR for colorizing LiDAR point cloud data is desired but not required.
- If integrated battery power is included, sensor provides for storage of 40 to 60 minutes of autonomous operation
- One (1) GPS antenna and cable

- Payload boresight calibrated and provide an angle correction certificate
- Minimum one (1) year warranty and maintenance on all hardware
- Perpetual software license for post-processing GNSS and IMU data (with future software updates included in base system package and with no annual license fees)
- Perpetual software license for data processing (with future software updates included in base system package and no annual license fees), including:
    - Correcting aircraft trajectories
    - Point cloud colorization and classification
    - Georeferencing
    - Strip adjustments
    - Merging of data from multiple flights
    - Export of data to LAZ/LAS formats
- Sensor firmware updates for life of the sensor
- Desktop software must be compatible with Windows 11 or higher
- Field case